

RITWIJ ARYAN PARMAR

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Available full-time immediately | MS completed May 2026

Software engineer building production ML systems across LLM serving runtimes, evaluation pipelines, and cloud-backed data workflows.

PROJECTS

HelixServe – LLM Serving Runtime

GitHub | Demo

- Built an LLM serving runtime on GCP NVIDIA L4 with paged KV cache, continuous batching, prefix caching, and CUDA Graph decode, increasing mixed-workload throughput from 175.8 to 1,007.8 tokens per second and reducing p95 end-to-end latency from 3.28 seconds to 0.80 seconds for production-style inference.
- Added inference benchmarking and observability for time-to-first-token, inter-token latency, throughput, and KV utilization, achieving 99.8 percent prefix hit rate and reducing repeated-prefix TTFT p95 from 0.299 seconds to 0.158 seconds.

ManoVarta – Controller-Led Multilingual Mental Health GenAI System

GitHub | Demo

- Engineered and deployed a 6-layer multilingual GenAI runtime on Google Cloud Run, integrating session orchestration, dialogue planning, LLM evidence extraction, structured scoring, and safety routing for English, Hindi, and Hinglish PHQ-9/GAD-7 assessments.
- Built the bilingual data and replay-evaluation pipeline with 60 dual-annotated sessions and 48 archived conversations, reaching 78.3 percent coverage completeness, 63.9 percent exact match, 0 parse failures, and 100 percent precision/recall on the evaluated safety set.

SRE-Nidaan – SRE Incident Response Copilot

GitHub | Demo

- Built and deployed a 3-microservice SRE incident response copilot on Google Cloud Run using Next.js, FastAPI, and vLLM, integrating 9 APIs for telemetry analysis, runbook retrieval, remediation gating, and analyst feedback.
- Aligned an LLM pipeline with QLoRA SFT, reward modeling, and RLHF-style data over 2,500 incident scenarios, adding audit trails and tenant-scoped reliability metrics for observable remediation workflows.

TECHNICAL EXPERIENCE

Networked Embedded Sensing Lab, Buffalo, NY

Jul 2025 – Feb 2026

Research Aide – Computer Vision and Infrastructure AI

- Built a culvert-barrel defect segmentation pipeline for inspection video, using SAM-3 text and exemplar prompts to mask crown cracks, wall spalling, joint separation, corrosion staining, standing water, and invert sediment across 1,400+ sampled frames.
- Cut SAM-3 inference by approximately 60 percent by sampling inspection video into stable barrel keyframes, dropping motion-blurred frames, removing near-duplicate wall/invert views, and propagating defect masks across adjacent frames with consistent defect IDs.
- Deployed the culvert-segmentation workflow on a Hugging Face GPU inference endpoint with a custom PyTorch container, SAM-3 artifacts, batch frame inputs, JSON mask/polygon outputs, endpoint logs, and per-run IoU/F1 reports, measuring approximately 0.74 mask IoU and 2.3 seconds p95 keyframe inference.

Tata Elxsi, Bengaluru, India

Jan 2024 – Jun 2024

Software Engineer Intern – ML Evaluation Systems

- Built a synthetic-data pipeline for perception model evaluation, synchronizing RGB, depth, semantic labels, pose, and calibration matrices to generate 2D/3D annotations for detector training and regression tests.
- Developed failure mining across 120+ night, rain, occlusion, cut-in, and pedestrian-crossing scenarios, improving held-out pedestrian/vehicle recall from 0.71 to 0.82 and reducing manual replay review by about 35 percent.

LLMate.ai, Bengaluru, India

Apr 2023 – Jul 2023

Backend Engineer Intern

- Built a governed LLM-to-SQL analytics layer over 50,000+ business records with schema retrieval, column pruning, SQL parser validation, read-only policy enforcement, row/time limits, and cost-aware query execution.
- Added dry-run checks, database-error tracing, clause-level repair, duplicate-request suppression, cache-backed result reuse, and 180+ replayable eval questions, reducing failed runs from about 31 percent to 12 percent and cutting repeated-request latency by about 40 percent.

SKILLS

Languages: Python, C++, Java, SQL

Machine Learning: PyTorch, Machine Learning, Deep Learning, Computer Vision, Object Detection, Model Evaluation, LLMs, RAG

Inference & MLOps: Model Serving, Model Deployment, Inference Optimization, vLLM, CUDA, GPU Computing, Docker

Backend & Cloud: FastAPI, Spring Boot, GCP, Cloud Run, BigQuery, RabbitMQ, Distributed Systems, Observability, CI/CD

EDUCATION

University at Buffalo – SUNY

Jan 2025 – May 2026

Master of Science in Computer Science and Engineering, GPA: 3.74 out of 4.0

Manipal Institute of Technology

Aug 2020 – Aug 2024

Bachelor of Technology in Computer Science and Engineering